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RNN Predication for COVID-19

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Abstract—**The global impact of the COVID-19 pandemic necessitates a comprehensive understanding of outbreak data for effective response. This study focuses on in-depth analysis using time series prediction models, specifically Long Short-Term Memory (LSTM) and Gated Recurrent Unit Network (GRUNet), applied to COVID-19 data from Hopkinson University. The study acknowledges the multifactorial nature of COVID-19 complexities, simplifying the problem to time-series data. The methods section details the mechanisms of RNNs, LSTM, and GRU, emphasizing their capabilities in capturing temporal dependencies and addressing long-term dependencies. The experimental phase visualizes COVID-19 data trends in the United States, highlighting the surge potentially linked to the Omicron variant. Two experimental models, LSTM and GRUNet, are compared, with training configurations and visualizations presented for each. However, the analysis of prediction results reveals limitations in accurately forecasting future COVID-19 trends, attributed to the dynamic nature of the virus and evolving strains, leading to increased transmissibility. The lack of effective control measures further hinders predictive capabilities.**

Keywords-COVID-19; LSTM and GRUNet; Time series analysis; Prediction models; Pandemic dynamics

I. INTRODUCTION

COVID-19 stands for "Coronavirus Disease-2019" and is a respiratory disease caused by a virus called Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2). SARS-CoV-2 belongs to the Coronaries family and is a single-stranded, positive-sense RNA virus that is highly contagious. highly infectious. It acts like an influenza virus, primarily affecting the respiratory system and causing symptoms including cough, fever, fatigue, and difficulty breathing. Although the exact origin of the virus is not yet known, scientists have determined that SARS-CoV-2 belongs to the β-CoV genus of the coronavirus family, commonly found in bats and rodents. Since COVID-19 was first reported in Wuhan, Hubei Province, China, in December 2019, the disease has spread rapidly to other parts of the world, with far-reaching consequences for human life. [1][2][3]

Deep learning algorithms excel in analyzing and forecasting outbreaks of diverse epidemics. When coupled with suitable intervention strategies, these methods frequently deliver accurate predictions for public health officials, aiding them in implementing timely interventions to contain the transmission of highly contagious diseases. In the research conducted by Natarajan et al., LSTM emerges as a promising tool for predicting COVID-19 pandemics through the assimilation of extensive data. Nevertheless, this study identified a limitation in the LSTM model, as it was unable to accurately capture the underlying patterns.[4] This study aims to study in-depth the COVID-19 data collected by Hopkinson University and explores the effect of different sequence lengths on network prediction by visualizing new cases in each country and training and fitting them with the help of LSTM and GRUNet time series prediction models. Specifically, we focus on the prediction of daily new cases in the United States and the trend of cumulative confirmed cases in the country. Using Russia as an example, we also provide insights into the difficulties faced in forecasting daily new and cumulative confirmed cases and provide a detailed analysis of the findings.

The global Impact of the COVID-19 is not only a health crisis, but also a serious challenge to social production and lifestyles [5]. Especially in the United States, the new crown epidemic has not slowed down since 2019 with the number of new infections approaching 1 million per day. [6] Therefore, accurate and reasonable prediction of the direction of the virus is of utmost significance and value for the effective organization of production and life.

In this study, we focus on the time-series data of new cases per day, recognizing that its complexity is affected by a combination of multiple factors, including viral infectivity, regional population, national preventive measures, vaccination status, and virus species. In our study, we simplify the problem to time-series data and apply a typical RNN for predictive solving. Through this study, we expect to provide strong support for the in-depth understanding of epidemic trends and a scientific basis for future defense and control decisions.

II. METHODS

A. RNN (Recurrent neural Network)

Recurrent Neural Networks (RNN) represent a class of artificial neural networks that focus on processing sequential data or time-series information. These deep learning algorithms are commonly used to tackle challenges that are sequential or temporal in nature, covering areas such as language translation, natural language processing (NLP), speech recognition, and image captioning. They have been embedded in a range of popular applications such as Siri, voice search, and Google Translate.

Unlike feedforward neural networks and convolutional neural networks (CNNs), recurrent neural networks improve

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performance by learning from training data. However, RNNs are unique in that they exhibit a special "memory function" that makes them ideal for time series analysis problems. Like how humans recall memories to deepen their understanding of the world, RNNs mimic this mechanism. Its main feature is its ability to retain a certain amount of memory in the processed information, which makes it excellent at processing temporally correlated data, whereas other neural network types usually lack this memory retention ability.

This unique ability allows RNNs to excel in processing complex natural language, speech, and image data, providing powerful solutions to a wide range of real-world problems. Its flexibility and adaptability make RNNs a powerful tool for solving a wide range of temporal tasks, driving the continued evolution of deep learning in multiple domains. [7]

RNNs employ neurons with self-feedback mechanisms, enabling them to process sequences of arbitrary lengths. Given an input sequence x1: T=x1, x2…, xt…, xT, the recurrent neural network updates the activity values of the hidden layer with feedback edges using the following formula:

$$\eta\tau=\phi\ (\eta\tau-1,\ \xi\tau) \tag{1}$$

where f is a function that defines the relationship between the previous hidden state ht-1 and the current input xt. This recurrent structure allows RNNs to effectively capture temporal dependencies in sequential data, making them a powerful tool for analyzing time series information.

B. LSTM (Long Short-Term Memory)

Long Short-Term Memory (LSTM) networks represent a unique form of Recurrent Neural Networks (RNNs) specifically designed to process sequential data and retain long-term dependencies. This network structure exhibits superior ability to process a wide variety of sequential data, including time series, text, and speech. Introducing the mechanism of storage units and gates to control the flow of information, LSTMs are able to selectively retain or discard information in a highly flexible manner, successfully overcoming the problem of gradient vanishing that is common in traditional RNNs. They have achieved wide success in a variety of fields such as natural language processing, speech recognition, and time series prediction.[8]

As a special type of recurrent neural networks (RNNs), LSTMs have made significant progress in solving the challenge of learning long-term dependencies and have further improved their performance through improvements and extensions by researchers such as Alex Graves. Their unique design aims to effectively address the problems associated with long-term dependencies. Unlike regular RNNs, LSTMs have the ability to automatically memorize long-term information by default, which is not a skill that requires great effort to acquire. This makes LSTMs a powerful tool for processing complex time-series data, providing reliable and efficient solutions for a variety of application scenarios. The widespread use of LSTMs in technology and artificial intelligence lays a solid foundation for future development.

Compared to traditional RNNs, LSTM exhibits superior performance in handling longer sequences. The key improvements of LSTM over the original RNN lie in two main aspects:

1) New Internal State

LSTM introduces a novel internal state $ht \in RD$ ct explicitly designed for linear cyclic information transfer. Simultaneously, it nonlinearly outputs information to the external state $ht \in RD$. The internal state ct is computed using the following formulas:

$$\chi\tau=\phi\tau\odot\chi\tau-1+\iota\tau\odot\chi\tau \tag{2}$$

$$\mathrm{ht} = \mathrm{ot} \odot \tan hct \tag{3}$$

Here, ft, it, and ot are three gates in the range [0,1] ^D that control the information flow pathways. The symbol ⊙ denotes element-wise multiplication. Ct-1 represents the memory cell from the previous time step, and ct∈RD is the candidate state obtained through a nonlinear function:

$$\chi\tau=\tau\alpha\nu\Omega\chi\tau+\Upsilon\chi\eta\tau-1+\beta\chi \tag{4}$$

At each time step t, the LSTM network's internal state ct records the historical information up to the current moment.

2) Gate Mechanism

In digital circuits, a gate is a binary variable $\{0, 1\}$. In the context of LSTM, this gate mechanism serves as a control for information flow. A value of 0 indicates a closed state, preventing any information from passing through, while a value of 1 signifies an open state, allowing all information to pass.

Through these enhancements, LSTM effectively tackles long-term dependency challenges and ensures robust performance in processing sequences of considerable length. This makes LSTM a powerful tool for various applications, especially in the context of time series analysis and prediction.

C. GRU (Gated Recurrent Unit)

The Gated Recurrent Unit (GRU) is recognized as an extremely effective variation of LSTM networks. Its appeal lies in its straightforward structure and impressive performance, making it a popular choice among neural network architectures. As a derivative of LSTM, GRU successfully addresses the challenge of handling long dependencies in RNN networks. While GRU and LSTM often show similar performance in various scenarios, GRU sets itself apart with its noteworthy computational efficiency.

What makes GRU distinctive is its streamlined internal structure, achieved by omitting one of the gating mechanisms. This reduction in complexity leads to a more efficient computation process. Despite this simplification, GRU maintains functionality comparable to LSTM. This computational efficiency makes GRU particularly attractive in situations where resource constraints or streamlined architectures are key considerations. [9][10]

The similarity in the input-output format between GRU and a regular RNN adds to its appeal. This similarity allows for easy integration of GRU into existing frameworks, reducing the learning curve for practitioners familiar with conventional RNN structures. Additionally, GRU shares conceptual foundations with LSTM, benefiting from the lessons learned and advancements made in LSTM architectures.

In essence, GRU's derivation from LSTM, combined with its simplicity, notable performance, and computational efficiency, positions it as a favored neural network architecture in contemporary settings. The versatility of GRU, seen in its structural compatibility with traditional RNNs and alignment with LSTM concepts, underscores its significance in the dynamic landscape of deep learning and neural network applications.

The updating formula for the hidden state (h') in GRU is expressed as follows:

$$\eta\tau=(1-\zeta)\ \odot\eta\tau-1+\zeta\odot\eta\backepsilon \qquad (5)$$

where $\odot$ represents element-wise multiplication, z is a gate controlling the flow of information, ht−1 is the previous hidden state, and h' is the updated hidden state. This formulation encapsulates the essence of the GRU mechanism, facilitating efficient information flow modulation in sequential data processing tasks.

III. Experiment and results

A. COVID-19 Data

Through the visualization of daily new data in the US, the blue line represents the daily addition of cases, while the red line represents the smoothed values (Fig. 1). It is evident that the daily additions exhibit oscillatory patterns, lacking smoothness and potentially featuring multiple peaks. Notably, towards the end of 2021, there is a significant surge in newly confirmed cases in the US, speculated to be influenced by the Omicron variant, which is known for its heightened transmissibility. This observation suggests a substantial impact of the Omicron variant on the transmission dynamics in the US.

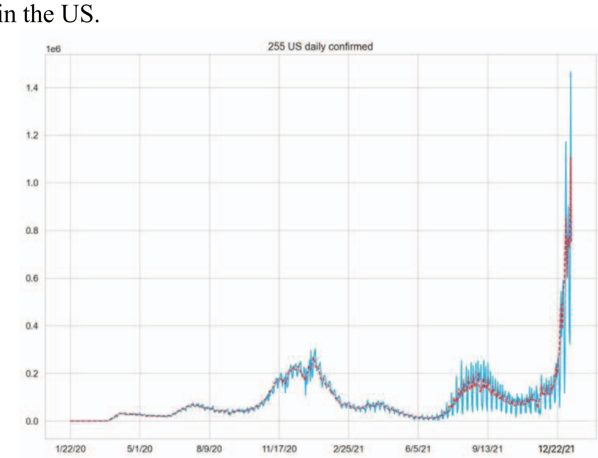

Fig. 1. 255 daily confirmed

B. Experimental Models

Two distinct models, namely Long Short-Term Memory (LSTM) and Gated Recurrent Unit Network (GRUNet), were employed for predictive analysis, and a comprehensive comparative evaluation was conducted.

1) LSTM Model

LSTM is widely recognized for its excellent results in predicting sequential data and has become a key component in our experimental framework. In order to ensure the reliability of our experimental results, we have carefully stratified the dataset into training and test sets that follow a specific ratio.

When applying the LSTM model for modeling, the length of the input data was set to a predetermined value, seq_length. This strategy aims to capture important patterns and dependencies in the data. In the training phase, we systematically input sequences of information into the LSTM model, with the length of each sequence matching the predetermined seq_length. This careful approach is intended to enable the model to learn complex temporal dependencies and subtle patterns in the data.

As the LSTM model is optimized during training, it becomes adept at recognizing and understanding the sequential nature of the input data. Once training is complete, we utilize the model to make predictions about the next data point based on a sequence of data whose length is equal to seq_length. This approach utilizes the LSTM's inherent memory and insights gained from the training data to make informed predictions about subsequent data points.

2) GRUNet Model

GRUNet is an innovative model designed to solve the long dependency problem in recurrent neural networks (RNNs) and is therefore included in our experimental design. Similar to the LSTM approach, we similarly divide the dataset into different training and testing subsets to ensure the reliability of the experiments. As a variant of LSTM, GRUNet exhibits similar performance in many cases, and is unique in that it is simpler in structure and more computationally efficient.

The introduction of this innovative GRUNet model enriches our experimental framework by providing us with another tool to deal with long dependencies. By employing different training and testing subsets, we ensure a comprehensive evaluation of the performance of this model. Compared to LSTM, GRUNet also performs well when dealing with sequential data, while its simpler structure and higher computational efficiency bring additional advantages to our experiments.

C. LSTM Prediction for the United States:

The LSTM model was employed to forecast trends in the United States. The model was trained over 1500 epochs using data specific to the United States. The initial learning rate was set to 1e-3, and a cosine decay strategy was implemented for learning rate adjustment. The Mean Squared Error (MSE) served as the designated loss function for training.

Training Configuration:

Epochs: 1500
Initial Learning Rate: 1e-3
Learning Rate Adjustment: Cosine Decay
Loss Function: Mean Squared Error (MSE)

The training process involved monitoring the decay of the learning rate (lr), tracking changes in the loss function, and generating visualizations depicting the learning rate decay, loss variation, and the model's prediction performance.

The figures below illustrate the learning rate decay, loss variation, and the predicted trends during the training process (Fig. 2):

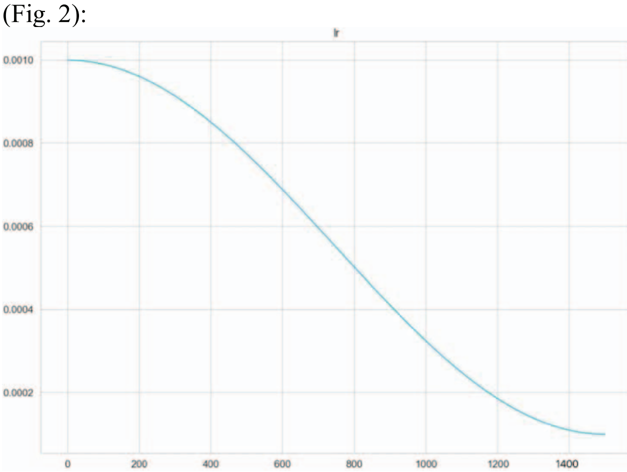

Fig. 2. Changes in learning rates

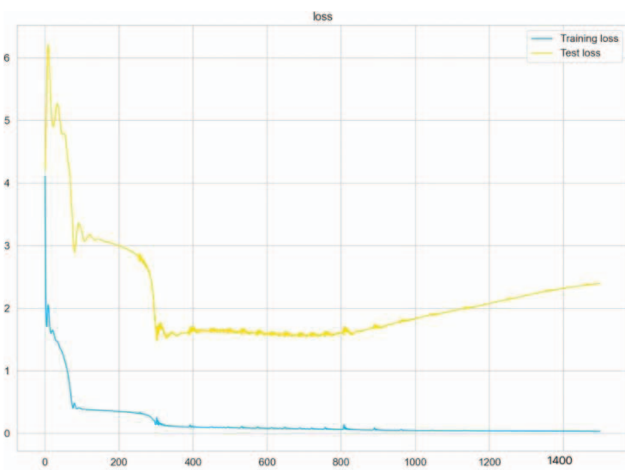

Fig. 3. Variation of loss during training using LSTM for US

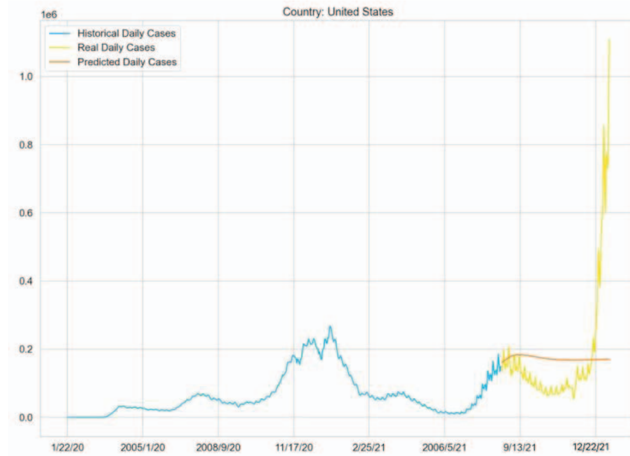

Fig. 4. Prediction of new daily outbreaks in the United States using LSTM

These visualizations are instrumental in comprehending the model's training dynamics and assessing its predictive capabilities for the United States dataset (Fig. 3 and Fig. 4).

D. GRUNet Prediction for the United States:

The training process involved tracking changes in the loss function and evaluating the predictive performance of the GRUNet model.

The results, including loss variation, and predictions, are presented in the figures below (Fig. 5):

Fig. 5. Variation of loss during training using GRUNet for US

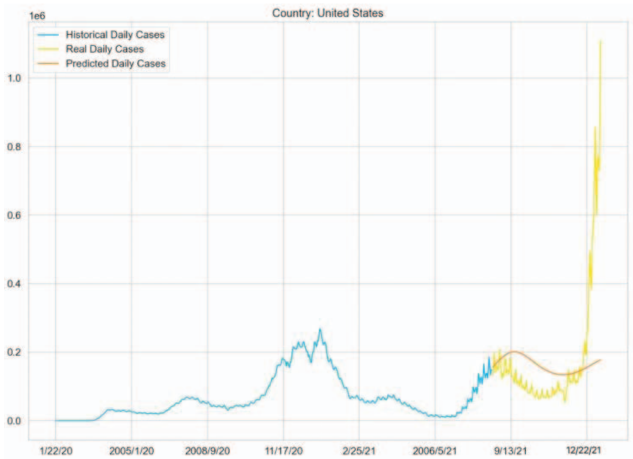

Fig. 6. Prediction of new daily outbreaks in the United States using GRUNet

These visualizations offer valuable insights into the training dynamics of the GRUNet model and its predictive capabilities for the United States dataset (Fig. 6).

E. Analysis of Prediction Results

A review of the projections found that the models did not show superior performance in predicting future new daily COVID-19 cases in the United States. This limitation is rooted in the evolutionary nature of the virus, with subsequent strains showing greater transmissibility, and this key information cannot be inferred from previous sequential data alone.

Unfortunately, the predictive performance of both models is affected by the dynamic and mutational characteristics of the virus. The lack of effectiveness of epidemic control measures in the United States, coupled with the unpredictability of virus mutation, makes it difficult for both models to accurately predict future trends.

This challenge highlights that relying solely on previous sequential data may not provide sufficient information when faced with highly mutated and dynamically evolving viruses. Therefore, more targeted, and adaptable epidemic control measures and more flexible predictive models may be the key to solving this problem. It also underscores the need to constantly adapt and improve our forecasting and containment strategies in response to evolving health crises.

IV. Conclusion

In the course of this investigation, the utilization of LSTM and GRUNet models for predicting the incidence of COVID-19 infections has provided valuable insights into the principles and applications of Recurrent Neural Networks (RNNs). The outcomes of these predictive models have been mixed, with some results proving satisfactory while others exhibited suboptimal effectiveness. This variance in performance underscores the complexity of forecasting the trajectory of the virus, signaling the necessity for the integration of additional influential factors into predictive models.

The ever-evolving nature of the virus poses a significant challenge to accurate forecasting, emphasizing the critical need for adaptable and well-informed public health strategies. The dynamic and unpredictable nature of the pandemic landscape requires strategies that can flexibly respond to changing circumstances. The observed limitations in the predictive capabilities of the LSTM and GRUNet models highlight the importance of ongoing research efforts and a commitment to continuous adaptation in addressing the dynamic nature of public health crises.

As we confront the challenges posed by the COVID-19 pandemic, it becomes evident that predictive modeling alone may not be sufficient to capture the full complexity of the virus's behavior. The integration of additional factors, such as socio-economic variables, healthcare infrastructure, and vaccination rates, is essential for enhancing the accuracy and reliability of predictive models. Furthermore, these findings underscore the need for interdisciplinary collaboration, data sharing, and a holistic approach to public health research to develop robust strategies for pandemic preparedness and response.

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